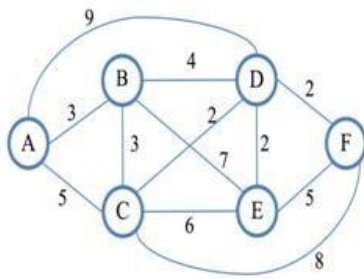


1. Find the shortest distance between vertices A and F using Dijkstra's algorithm for the given graph and draw the path.

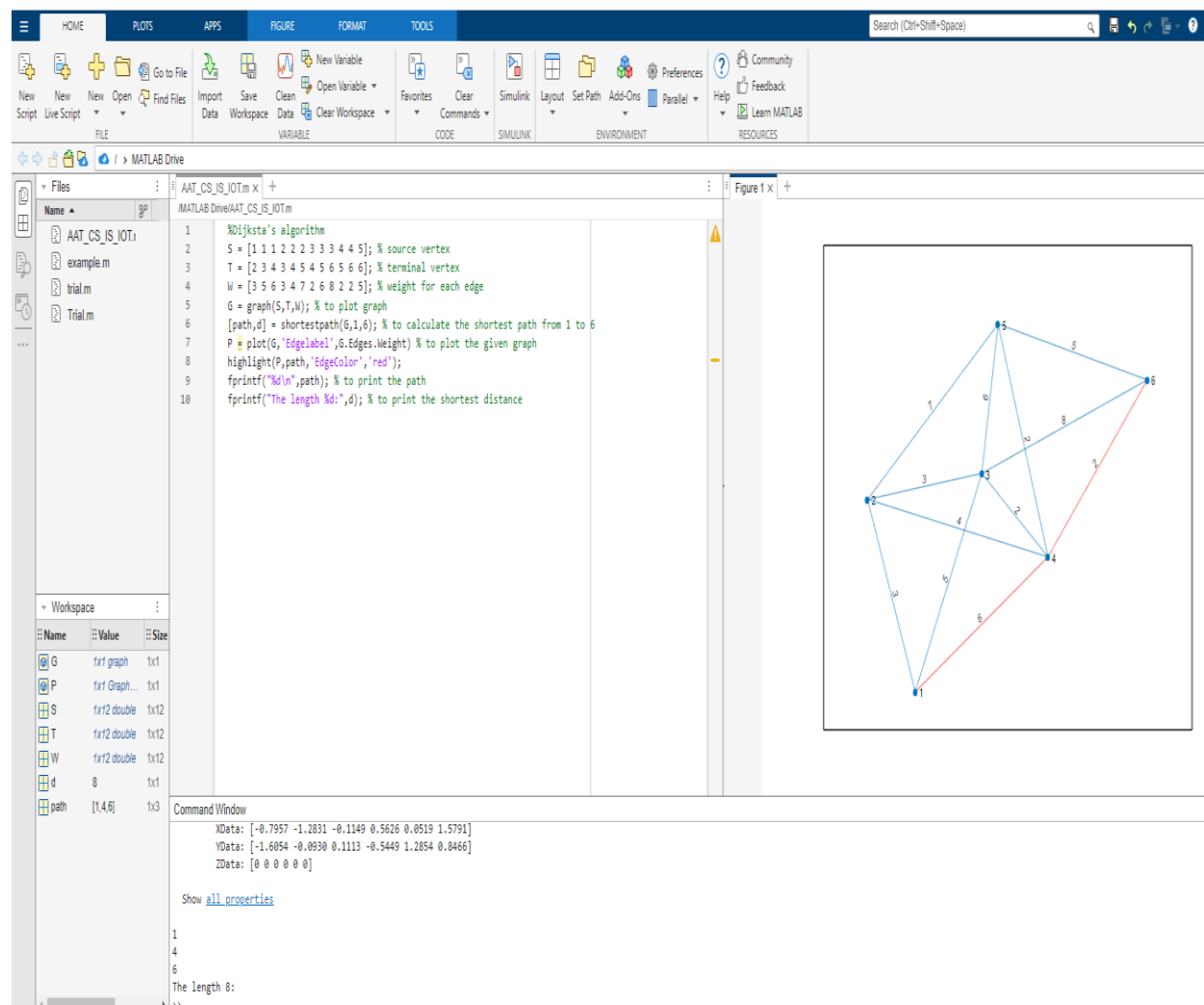


%Dijkstra's algorithm

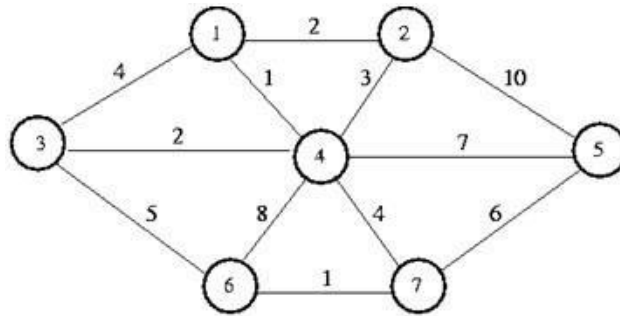
```

S = [1 1 1 2 2 2 3 3 3 4 4 5]; % source vertex
T = [2 3 4 3 4 5 4 5 6 5 6 6]; % terminal vertex
W = [3 5 6 3 4 7 2 6 8 2 2 5]; % weight for each edge
G = graph(S,T,W); % to plot graph
[path,d] = shortestpath(G,1,6); % to calculate the shortest path from 1 to 6
P = plot(G,'Edgelabel',G.Edges.Weight) % to plot the given graph
highlight(P,path,'EdgeColor','red');
fprintf("%d\n",path); % to print the path
fprintf("The length %d:",d); % to print the shortest distance
  
```

Output:

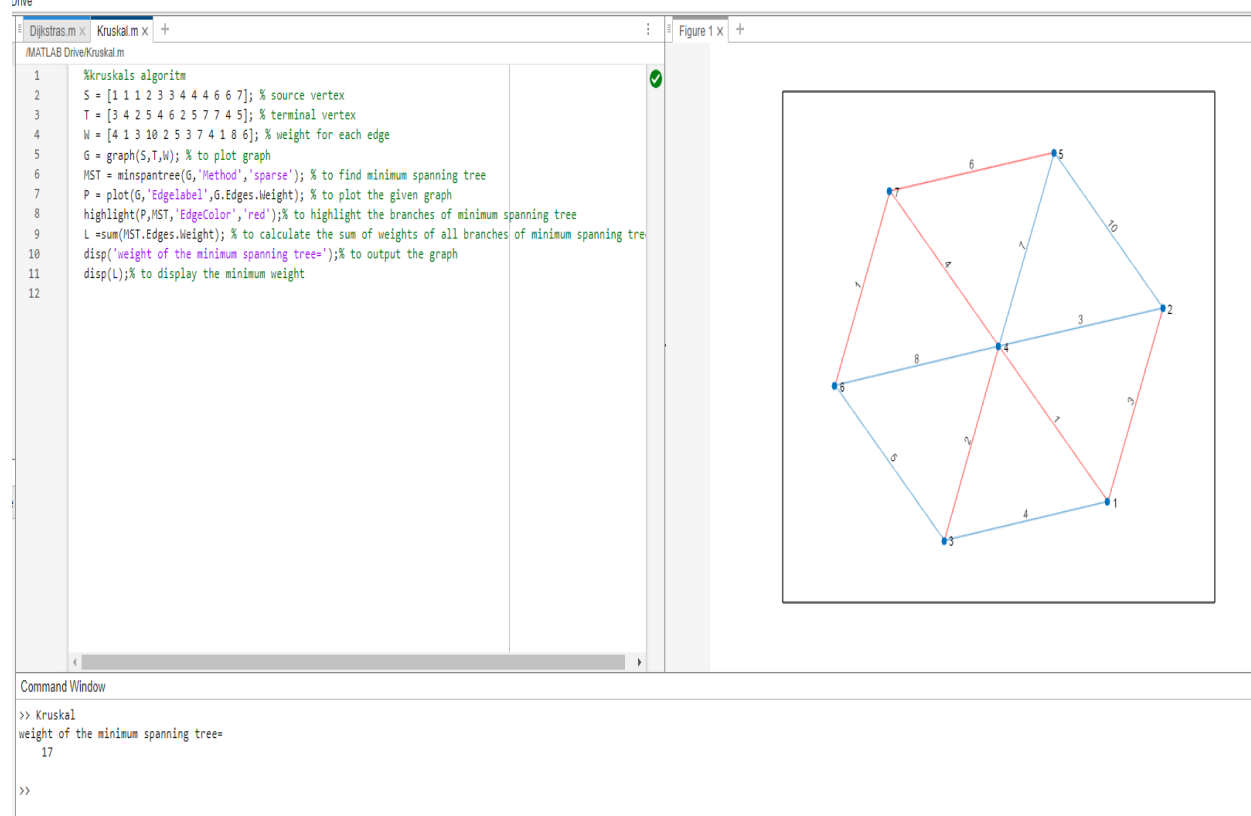


2. Find the minimum spanning tree of the given graph using Kruskal's algorithm and write its weight.



```
%kruskals algorithm
S = [1 1 1 2 3 3 4 4 4 6 6 7]; % source vertex
T = [3 4 2 5 4 6 2 5 7 7 4 5]; % terminal vertex
W = [4 1 3 10 2 5 3 7 4 1 8 6]; % weight for each edge
G = graph(S,T,W); % to plot graph
MST = minspantree(G,'Method','sparse'); % to find minimum spanning tree
P = plot(G,'Edgelabel',G.Edges.Weight); % to plot the given graph
highlight(P,MST,'EdgeColor','red'); % to highlight the branches of minimum spanning tree
L =sum(MST.Edges.Weight); % to calculate the sum of weights of all branches of minimum spanning tree
disp('weight of the minimum spanning tree='); % to output the graph
disp(L); % to display the minimum weight
```

Output:



3. The number of particles of contamination that occur on an optical disc has a Poisson distribution, and the average number of particles is 10. Find the probability that

(i) 12 or less particles occur in the area of a disc.

(ii) more than 5 particles occur in the area of a disc.

(iii) 3 particles occur in the area of a disc.

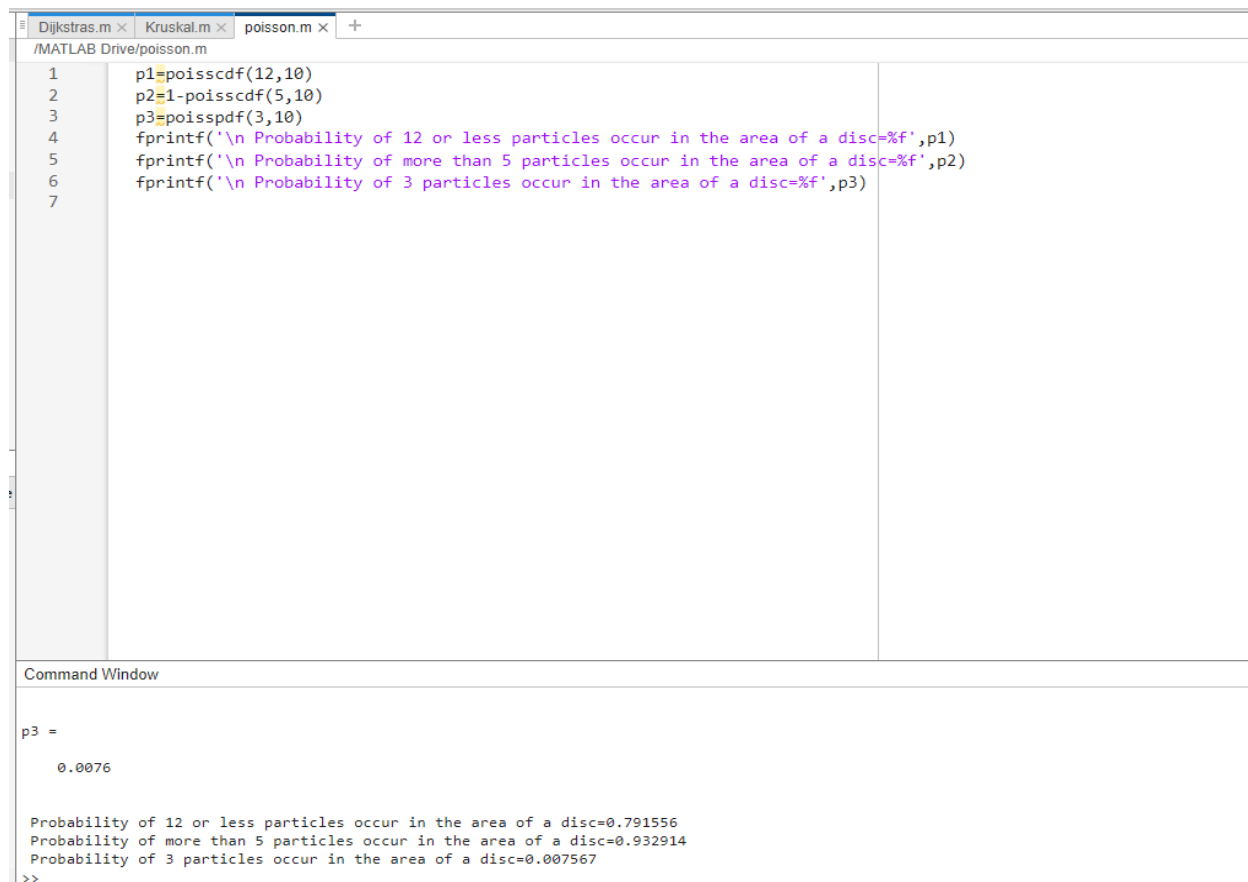
Syntax:

P1=poisscdf(x,lambda) (This syntax will calculate the probability for less than or equal to x)

P2=poisspdf(x,lambda) (This will evaluate at the point x)

Solution:

```
p1=poisscdf(12,10)
p2=1-poisscdf(5,10)
p3=poisspdf(3,10)
fprintf('\n Probability of 12 or less particles occur in the area of a disc=%f',p1)
fprintf('\n Probability of more than 5 particles occur in the area of a disc=%f',p2)
fprintf('\n Probability of 3 particles occur in the area of a disc=%f',p3)
```

A screenshot of the MATLAB environment. The top part shows the MATLAB Editor with three tabs: 'Dijkstras.m', 'Kruskal.m', and 'poisson.m'. The 'poisson.m' tab is active, showing a script with 7 lines of code. The code calculates the probability of 12 or less particles (p1), the probability of more than 5 particles (p2), and the probability of 3 particles (p3) using Poisson distribution functions. The bottom part shows the Command Window, which displays the output of the script. It shows the value of p3 as 0.0076, followed by three lines of formatted output: 'Probability of 12 or less particles occur in the area of a disc=0.791556', 'Probability of more than 5 particles occur in the area of a disc=0.932914', and 'Probability of 3 particles occur in the area of a disc=0.007567'. The Command Window ends with a prompt '>>'.

```
Dijkstras.m x Kruskal.m x poisson.m x +
/MATLAB Drive/poisson.m
1 p1=poisscdf(12,10)
2 p2=1-poisscdf(5,10)
3 p3=poisspdf(3,10)
4 fprintf('\n Probability of 12 or less particles occur in the area of a disc=%f',p1)
5 fprintf('\n Probability of more than 5 particles occur in the area of a disc=%f',p2)
6 fprintf('\n Probability of 3 particles occur in the area of a disc=%f',p3)
7

Command Window

p3 =
    0.0076

Probability of 12 or less particles occur in the area of a disc=0.791556
Probability of more than 5 particles occur in the area of a disc=0.932914
Probability of 3 particles occur in the area of a disc=0.007567
>>
```

4. The marks of 1000 students in an examination follows a normal distribution with mean 70 and standard deviation 5. Find

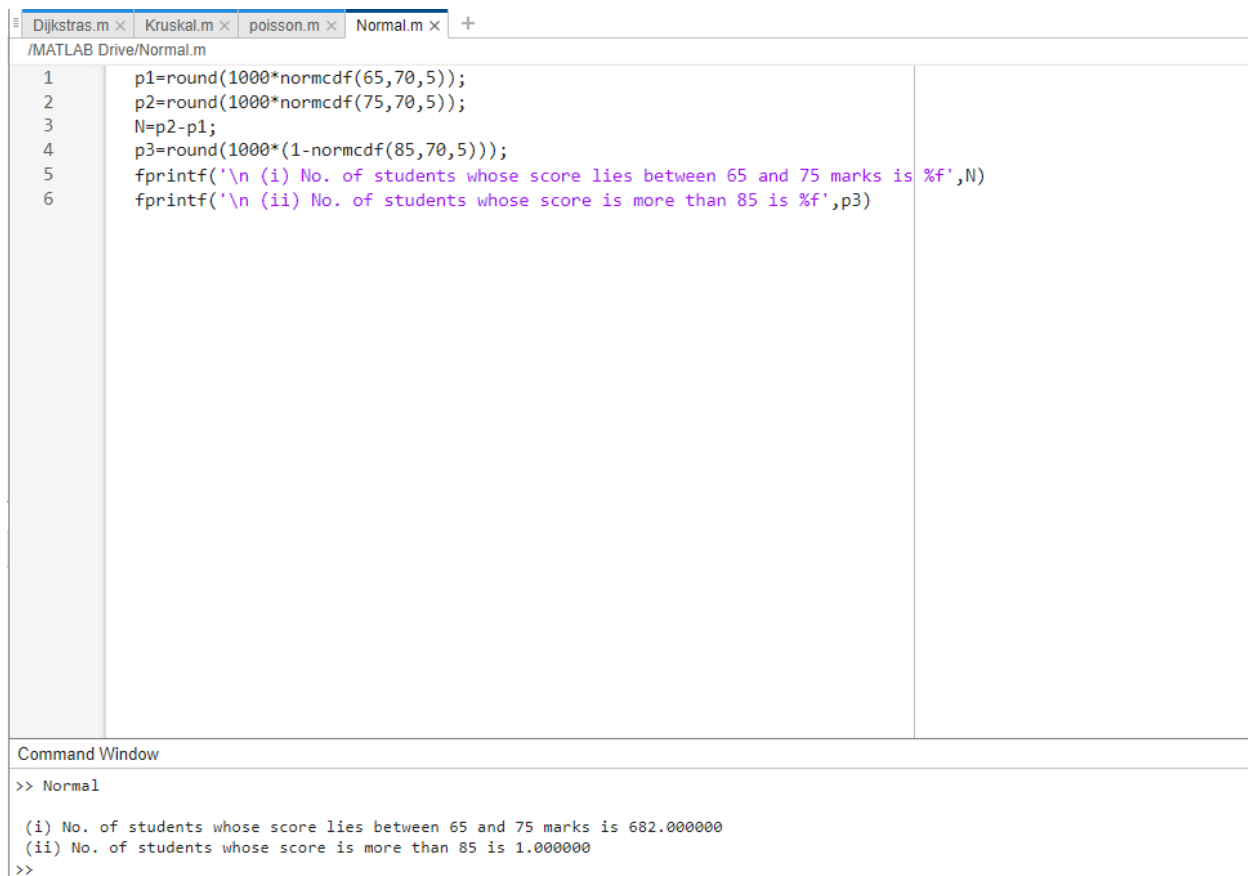
(i) no. of students whose score lies between 65 and 75 marks.

(ii) no. of students whose score more than 85.

P = normcdf(x, mu, sigma) (This syntax will calculate the probability for less than or equal to x)

Solution:

```
p1=round(1000*normcdf(65,70,5));
p2=round(1000*normcdf(75,70,5));
N=p2-p1;
p3=round(1000*(1-normcdf(85,70,5)));
fprintf('\n (i) No. of students whose score lies between 65 and 75 marks is %f',N)
fprintf('\n (ii) No. of students whose score is more than 85 is %f',p3)
```



The image shows a MATLAB environment with a script editor and a command window. The script editor has four tabs: 'Dijkstras.m', 'Kruskal.m', 'poisson.m', and 'Normal.m'. The 'Normal.m' tab is active, showing the following code:

```
1 p1=round(1000*normcdf(65,70,5));
2 p2=round(1000*normcdf(75,70,5));
3 N=p2-p1;
4 p3=round(1000*(1-normcdf(85,70,5)));
5 fprintf('\n (i) No. of students whose score lies between 65 and 75 marks is %f',N)
6 fprintf('\n (ii) No. of students whose score is more than 85 is %f',p3)
```

The command window shows the output of the script:

```
>> Normal

(i) No. of students whose score lies between 65 and 75 marks is 682.000000
(ii) No. of students whose score is more than 85 is 1.000000
>>
```

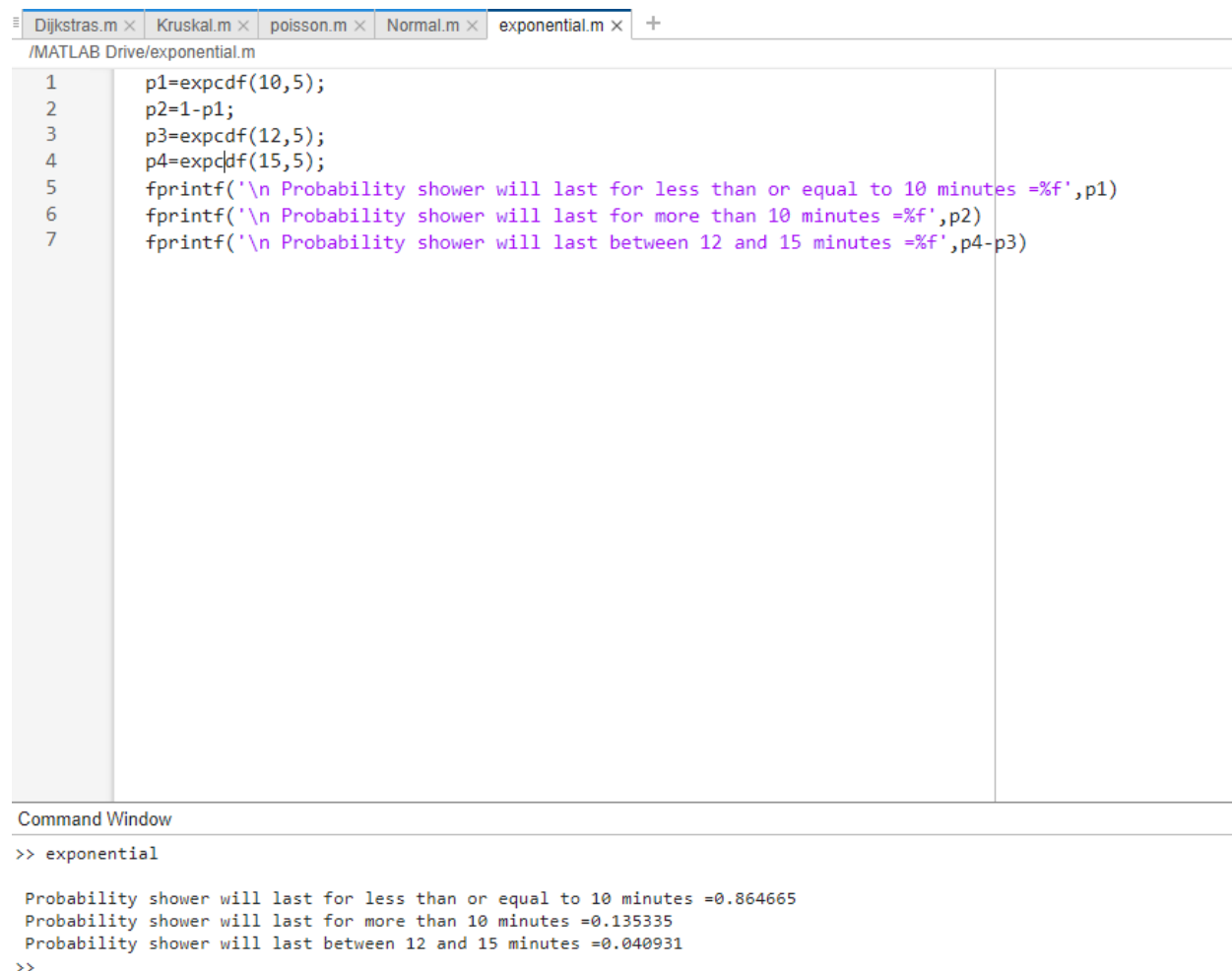
5. In a certain town, the duration of shower is exponentially distributed with mean 5 minutes. What is the probability that the shower will last for (i) less than or equal to 10 minutes (ii) more than 10 minutes and (iii) between 12 and 15 minutes?

Syntax:

P=expcdf(x,alpha)) (This syntax will calculate the probability for less than or equal to x)

Solution:

```
p1=expcdf(10,5);
p2=1-p1;
p3=expcdf(12,5);
p4=expcdf(15,5);
fprintf('\n Probability shower will last for less than or equal to 10 minutes
=%f',p1)
fprintf('\n Probability shower will last for more than 10 minutes =%f',p2)
fprintf('\n Probability shower will last between 12 and 15 minutes =%f',p4-p3)
```



The image shows a MATLAB editor window with the file 'exponential.m' open. The code in the editor is as follows:

```
1 p1=expcdf(10,5);
2 p2=1-p1;
3 p3=expcdf(12,5);
4 p4=expcdf(15,5);
5 fprintf('\n Probability shower will last for less than or equal to 10 minutes =%f',p1)
6 fprintf('\n Probability shower will last for more than 10 minutes =%f',p2)
7 fprintf('\n Probability shower will last between 12 and 15 minutes =%f',p4-p3)
```

Below the editor is the Command Window, which shows the output of the script:

```
>> exponential

Probability shower will last for less than or equal to 10 minutes =0.864665
Probability shower will last for more than 10 minutes =0.135335
Probability shower will last between 12 and 15 minutes =0.040931
>>
```

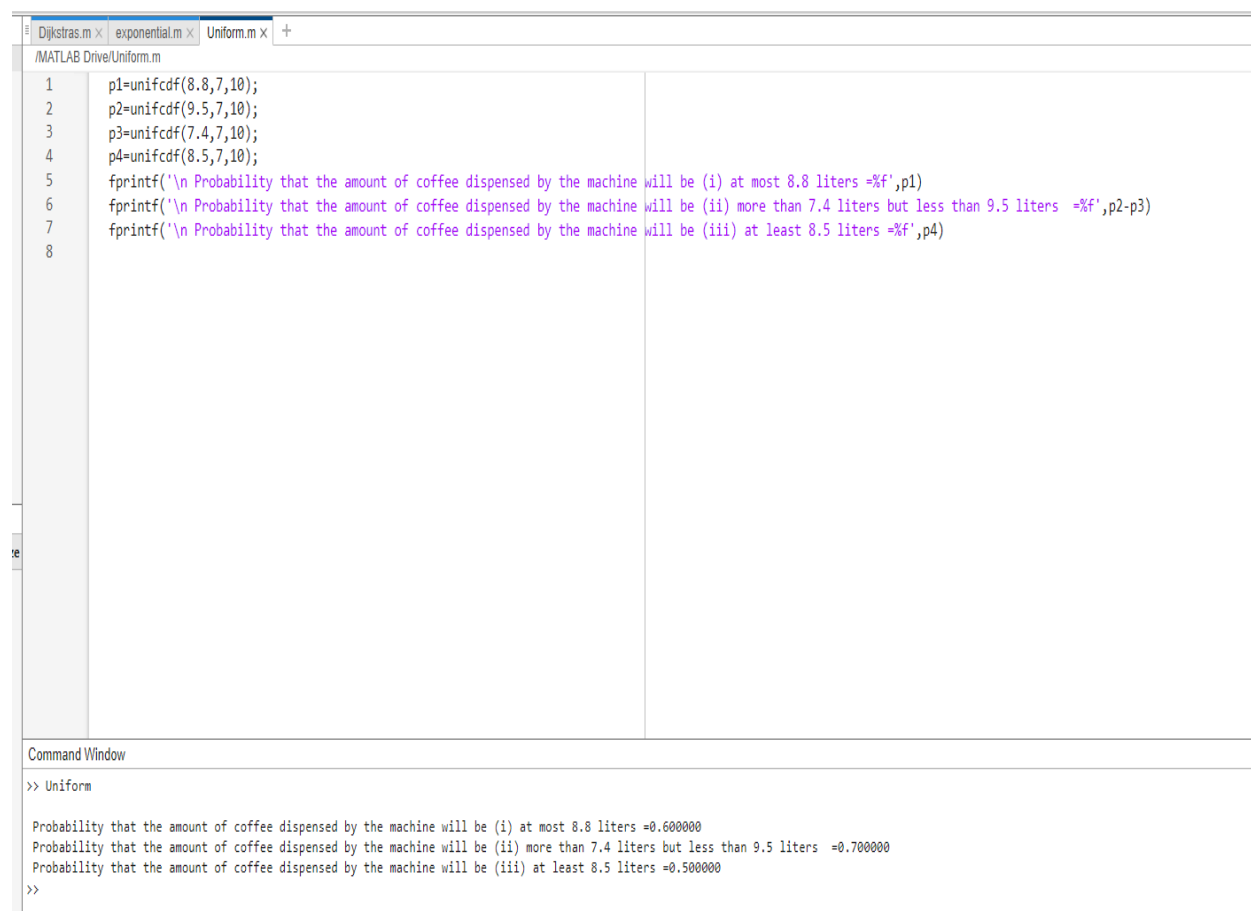
6. The daily amount of coffee (in liters) dispensed by a machine is uniformly distributed with $a = 7$ and $b = 10$. Determine the probability that the amount of coffee dispensed by the machine will be (i) at most 8.8 liters (ii) more than 7.4 liters but less than 9.5 liters (iii) at least 8.5 liters.

Syntax:

P=unifcdf(x,a,b) (This syntax will calculate the probability for less than or equal to x)

Solution:

```
p1=unifcdf(8.8,7,10);
p2=unifcdf(9.5,7,10);
p3=unifcdf(7.4,7,10);
p4=unifcdf(8.5,7,10);
fprintf('\n Probability that the amount of coffee dispensed by the machine will be
(i) at most 8.8 liters =%f',p1)
fprintf('\n Probability that the amount of coffee dispensed by the machine will be
(ii) more than 7.4 liters but less than 9.5 liters =%f',p2-p3)
fprintf('\n Probability that the amount of coffee dispensed by the machine will be
(iii) at least 8.5 liters =%f',p4)
```



The screenshot shows a MATLAB environment with three tabs: 'Dijkstra.m', 'exponential.m', and 'Uniform.m'. The 'Uniform.m' tab is active, displaying the following code:

```
1 p1=unifcdf(8.8,7,10);
2 p2=unifcdf(9.5,7,10);
3 p3=unifcdf(7.4,7,10);
4 p4=unifcdf(8.5,7,10);
5 fprintf('\n Probability that the amount of coffee dispensed by the machine will be (i) at most 8.8 liters =%f',p1)
6 fprintf('\n Probability that the amount of coffee dispensed by the machine will be (ii) more than 7.4 liters but less than 9.5 liters =%f',p2-p3)
7 fprintf('\n Probability that the amount of coffee dispensed by the machine will be (iii) at least 8.5 liters =%f',p4)
8
```

Below the code editor is the 'Command Window' showing the output of the script:

```
>> Uniform

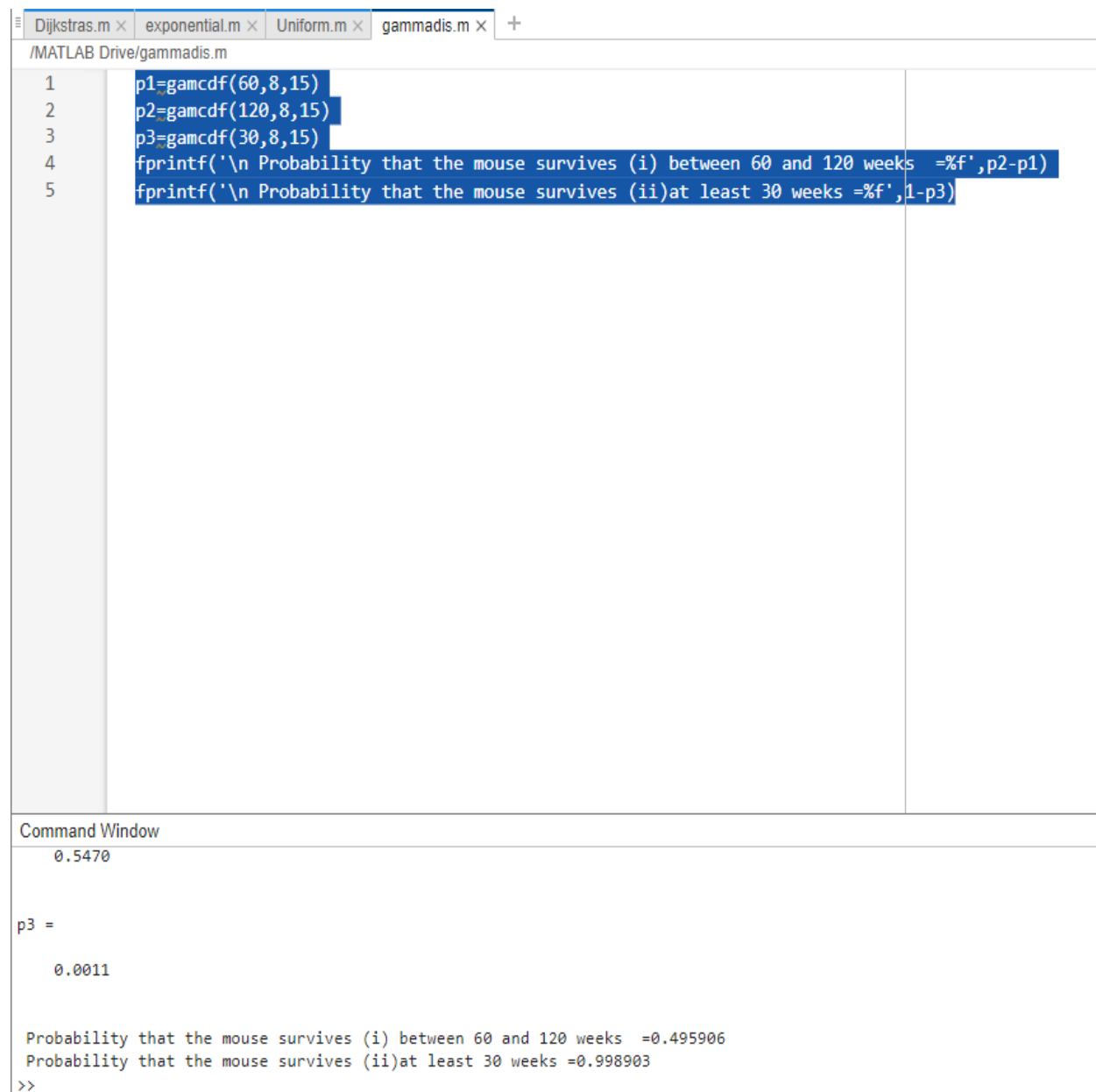
Probability that the amount of coffee dispensed by the machine will be (i) at most 8.8 liters =0.600000
Probability that the amount of coffee dispensed by the machine will be (ii) more than 7.4 liters but less than 9.5 liters =0.700000
Probability that the amount of coffee dispensed by the machine will be (iii) at least 8.5 liters =0.500000
>>
```

7. The survival time (in weeks) of a male mouse exposed to radiation has a gamma distribution with $\alpha = 8, \beta = 15$. Find the probability that the mouse survives (i) between 60 and 120 weeks (ii) at least 30 weeks.

Syntax:

gamcdf(x,a,b) (This syntax will calculate the probability for less than or equal to x)

```
p1=gamcdf(60,8,15)
p2=gamcdf(120,8,15)
p3=gamcdf(30,8,15)
fprintf('\n Probability that the mouse survives (i) between 60 and 120 weeks
=%f',p2-p1)
fprintf('\n Probability that the mouse survives (ii)at least 30 weeks =%f',1-p3)
```

A screenshot of the MATLAB environment. The top part shows the MATLAB Editor with a file named 'gammadis.m' open. The editor contains five lines of MATLAB code: p1=gamcdf(60,8,15); p2=gamcdf(120,8,15); p3=gamcdf(30,8,15); fprintf('\n Probability that the mouse survives (i) between 60 and 120 weeks =%f',p2-p1); and fprintf('\n Probability that the mouse survives (ii)at least 30 weeks =%f',1-p3). The bottom part shows the Command Window. It displays the output of the code: 0.5470, p3 = 0.0011, and two lines of formatted text: 'Probability that the mouse survives (i) between 60 and 120 weeks =0.495906' and 'Probability that the mouse survives (ii)at least 30 weeks =0.998903'. The Command Window prompt is '>>'.

```
Dijkstras.m × exponential.m × Uniform.m × gammadis.m × +
/MATLAB Drive/gammadis.m
1 p1=gamcdf(60,8,15)
2 p2=gamcdf(120,8,15)
3 p3=gamcdf(30,8,15)
4 fprintf('\n Probability that the mouse survives (i) between 60 and 120 weeks =%f',p2-p1)
5 fprintf('\n Probability that the mouse survives (ii)at least 30 weeks =%f',1-p3)

Command Window
0.5470

p3 =
0.0011

Probability that the mouse survives (i) between 60 and 120 weeks =0.495906
Probability that the mouse survives (ii)at least 30 weeks =0.998903
>>
```

8. An old lawn mower has a 20% chance of starting on a particular pull. Find the probability that it takes
(i) exactly 3 pulls to start the mower (ii) 10 or fewer pulls to start the mower.

Syntax:

P1=geocdf(x,p) (This syntax will calculate the probability for less than or equal to x)

P2=geopdf(x,p) (This will evaluate at the point x)

```
x1 = 3;  
x2 = 10;  
p = 0.2;  
p1 = geopdf(x1,p);  
p2 = geocdf(x2,p);  
fprintf('\n Probability that exactly 3 pulls to start the mower =%f',p1)  
fprintf('\n Probability that 10 or fewer pulls to start the mower =%f',p2)
```

